

JONATHAN THANGADURAI SELVARAJ

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Software and data professional with experience in systems integration, IoT pipelines, and applied AI workflows. Built industrial data integrations and monitoring systems, translating operational needs into usable technical solutions in collaboration with specialists and end users.

EXPERIENCE

TNO - Netherlands Organisation for Applied Scientific Research

Traineeship

Sept 2024 – Present

The Hague, Netherlands

Product Owner - Intelligence Pipeline

Jan 2026 – Present

- Proposed and led development of an **LLM-based pipeline** that converts corporate disclosures into structured, source-traceable intelligence; **piloted and now used by the Market Strategy team**.
- Designed the **Pydantic extraction schema**, validation contract, and provenance model, collaborating with data scientists to test and refine source-grounded extraction across complex corporate reports.
- Developed the **Azure infrastructure cost model** and scaling roadmap, aligning stakeholders across five internal functions on broader adoption within the organisation.

Project Manager - Air Quality & Emissions

May 2025 – Dec 2025

- Coordinated delivery across **five applied-research projects** in air quality and emissions, working with researchers, industry partners, and government organisations.
- Built a cost-modelling capability using Python scripts to turn compute-usage data into structured Excel estimates for scoping follow-on project funding.

Organisation Strategy Consultant - Geological Survey of the Netherlands

Sept 2024 – Apr 2025

- Worked with the **Enterprise Architecture** team to map products, data sources, dependencies, and accountable owners across a six-department division.
- Findings informed how the division could adopt a **product-oriented** way of working and organise teams **across the value chain** to tackle siloed handoffs between teams.

Solita – Industry Capstone

Apr 2023 – Jul 2023

Project Lead & Developer

Lappeenranta, Finland

- Led a **five-person team** in designing **APIs** and traceable **data-sharing flows** for a data-space prototype integrating material and waste data from multiple mining-sector providers.

Flex Ltd.

Oct 2020 – Jul 2022

Integration Solutions Consultant

Chennai, India

- Built and supported **API-based integration flows** handling production data across factories in India and Southeast Asia, capturing production, quality-control, and error-log data and making it **available to data analysts** for downstream processing and optimisation.
- Contributed to Flex's digital-transformation programme by developing and deploying new **MuleSoft integration** patterns between factory systems and global ERP platforms, using buffered feeds and scheduled processing to handle operational data reliably at scale.

RELEVANT PROJECTS

VaxiChain – IoT Cold-Chain Monitoring Platform

Nov 2022 – Feb 2023

- Built a simulated **real-time monitoring and alerting system** for vaccine cold-chain conditions: sensor data ingested over **MQTT**, using **OpenHAB** to trigger automated alerts when temperature readings went out of range.

RFID-Based Toll Collection System

May 2019 – Mar 2020

- Built an **RFID-based tolling** prototype for road networks: integrated a UHF **RFID** reader on **Raspberry Pi**, through serial communications and publishing them over **MQTT** to a cloud-based pub/sub service.

EDUCATION

Erasmus Mundus Joint Master's Degree (EMJMD) - SE4GD

Sept 2022 – Aug 2024

MSc Software Engineering & Sustainability

Europe

VU Amsterdam (NL) · LUT University (FI) · University of L'Aquila (IT)

Anna University

Jul 2016 – Oct 2020

BTech Information Technology

Chennai, India

TECHNICAL SKILLS

Core: Python · SQL · REST APIs · MuleSoft · Data modelling · Systems integration

Applied AI & Data: GenAI workflow design · Pydantic · Schema design · Data validation · Data provenance

Cloud, Databases & IoT: Azure · PostgreSQL · InfluxDB · MQTT · Grafana · Redis · Telegraf · Docker

Additional: Git · OpenHAB · Kotlin · JavaScript/TypeScript · HTML/CSS · Product Development ·